

Gabriel Bridger

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Summary

Computer Science with Artificial Intelligence student with a strong interest in applying AI and machine learning into financial markets. Skilled in time series forecasting & currently seeking an internship in Quantitative Research.

Education

University of Nottingham

June 2028

MSc Computer Science with AI - 4 Year Integrated Masters Course

First Year Grade Average: **78%**

Relevant Courses:

Functional Programming in Haskell, Fundamentals of Artificial Intelligence, **Developing Trading Algorithms**, Mathematics for Computer Scientists - **Advanced Linear Algebra** and Applications.

Relevant Topics:

RNNs, ANNs, LSTMs, ARIMA models, Time Series Analysis, Large Scale Data Processing, Discrete Dynamical Systems, Stochastic Matrices & Processes, Probability theory.

Work Experience & Projects

NEFS Quant | Quantitative Research Analyst

September 2025 - Present

- Selected as 1 of 14 from 200+ applicants, developing a strong conceptual understanding of quantitative finance.
- Competing in market-making games while developing skills in probabilistic thinking, game theory and quick mental maths.

Quantitative models for Time Series Forecasting | *Python, Shell, Scikit-Learn, XGBoost, Numpy, Pandas*

- Developed and compared multiple models to forecast stock returns, including an **XGBoost** implementation and more recently a **Markov-Switching GARCH** (MS-AR-GARCH) model built from scratch
- Engineered a parallelised **Recursive Feature Elimination** pipeline to optimise the input features
- Implemented a hybrid strategy using a **Monte Carlo Simulation** to produce **Conditional Value at Risk** metrics which allows the algorithm to determine stock position based on the level of acceptable risk
- Architected a containerised backtesting software to enable rapid and efficient model evaluation

Interactive Options Pricing Heatmap | *Python, Matplotlib/Seaborn*

- Built a UI to generate heatmaps of option prices calculated with the **Black-Scholes model**.
- The tool visualizes the value of call and put options, allowing for dynamic user inputs for variables like underlying price, strike price, volatility, risk-free rate, and time to expiration.

JPMorgan Chase & Co. Quantitative Research Virtual Experience Program on Forage - September 2025

- Analyzed a book of loans to estimate a customer's probability of default
- Used dynamic programming to convert FICO scores into categorical data to predict defaults

Skills

- **Languages:** Python, Haskell, SQL, C, Git
- **Libraries/Frameworks:** TensorFlow/PyTorch, Scikit-Learn, Pandas, NumPy, Matplotlib, yfinance, Linux
- **Quantitative Concepts:** Time Series Analysis (LSTMs, RNNs, ANNs), Black-Scholes, Monte Carlo, CVaR, Stochastic Processes